

Conditional Forecasting with Uncertainty: Scenarios, Soft Conditions, and Historical Counterfactuals in a Monetary VAR

A methods-and-tooling companion

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Built on the Canadian monetary VAR of Moran, Stevanovic & Surprenant (BoC SWP 2025-28)

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Abstract

Central-bank risk-scenario exercises are typically reported as point paths: “if oil rises to \$120, inflation does this.” The underlying Waggoner–Zha (WZ) conditional-forecast machinery, however, is fundamentally a statement about a *distribution* of shocks consistent with a scenario, and it therefore carries its own measure of uncertainty that is almost never shown. This paper presents an operational toolkit that puts that uncertainty front and centre. Working in a reduced-form vector autoregression identified recursively, we (i) derive the conditional-forecast *fan chart*, decomposing its width into a closed-form *shock* (innovation) component and a simulated *parameter* (estimation) component, and show that the constrained variable’s band collapses to zero by construction; (ii) formalize *soft* conditions, which condition on a distribution for the constrained variable rather than an exact path, so that the constrained variable itself acquires a band; (iii) treat *joint* and *partial-horizon* scenarios as nothing more than extra rows of one linear restriction; and (iv) derive *historical counterfactuals* as a minimum-norm perturbation of realized history, optionally confined to a chosen structural shock. Each object is stated with the intuition and the exact algebra implemented in the accompanying code, with full derivations in the appendix. The toolkit is delivered as a reproducible offline pipeline and a self-contained interactive browser dashboard in which the impulse responses, scenario fan charts, and counterfactuals are recomputed live. The Canadian monetary VAR of Moran, Stevanovic & Surprenant (2025) serves as the running validation case.

1 Introduction

Scenario analysis is a staple of policy institutions: condition the model on a hypothetical path for one or more variables and read off what the rest of the economy does. The dominant technical device is the Waggoner–Zha (1999) conditional forecast, which finds the “most likely” sequence of future shocks consistent with the imposed path and propagates it to every variable. Moran, Stevanovic & Surprenant (2025, hereafter MSS) apply exactly this device to a monthly vector autoregression (VAR) for the Canadian economy and report four risk scenarios. Those exercises, like most of their kind, report conditional *means*. Yet Waggoner and Zha’s original contribution is largely about the *distribution* around those means. A scenario is produced by shocks, and the shocks it does not pin down remain random, so the conditional forecast of every unconstrained variable is itself a random object with a well-defined predictive distribution. Reporting only its mean discards precisely the information a risk exercise is meant to convey—how uncertain the consequences of the scenario are.

This paper is a methods-and-tooling companion that makes conditional forecasting operational *with* its uncertainty. Its central object is the conditional-forecast fan chart, whose width we derive and decompose into two additive sources: a *shock* component—the innovations the constraint leaves free, available in closed form—and a *parameter* component—the estimation uncertainty in the VAR, obtained by re-solving the scenario over posterior or bootstrap draws of the coefficients. A by-product of the derivation, visible in every figure below, is that the constrained variable’s band collapses to a point by construction: a scenario pins what it pins with certainty and lets uncertainty accumulate only where the model is free to move. Around this core we develop the vocabulary a working risk desk actually needs. Soft conditions replace an exact path by a distribution—“oil around \$120, give or take”—so that the constrained variable, too, carries a band; joint and partial-horizon scenarios, which constrain several variables at once or a variable over only part of the horizon, turn out to be nothing more than adding or deleting rows of a single linear restriction; and historical counterfactuals—“what would inflation have done had the Bank tightened 100 basis points more?”—emerge as a minimum-norm perturbation of realized history, optionally confined to a single structural shock. Each object is stated with its intuition and the exact algebra implemented in the accompanying code, and derived in full in the appendices.

The apparatus is delivered as software, and the delivery is part of the point. The estimation runs offline from a cached panel, so a result is deterministic and archivable, and it is guarded by a regression suite that checks the algebraic invariants any correct implementation must satisfy. A single self-contained browser dashboard then exposes the toolkit interactively: the impulse responses, the scenario fan charts with their live shock band, and the counterfactuals are all re-computed in the browser from the exported model objects, with no server and no re-run of the estimation. Throughout we take the estimated VAR and its recursive identification from MSS as given and validated; the reduced-form model, the baseline forecast, the conditional-forecast mean, and the variance decomposition are reproduced, for completeness, in Appendix A, and the new derivations—the uncertainty decomposition, soft and joint conditions, and counterfactuals—are collected in Appendix B. The Canadian monetary VAR of MSS is the running example, and the figures are those produced by the pipeline and archived in the HTML report that accompanies the project.

Notation is kept light. $y_t \in \mathbb{R}^K$ is the vector of endogenous variables, p the lag order, H the horizon, and $\mathcal{F}_t = \{y_s\}_{s \leq t}$ the information set. Reduced-form innovations u_t have mean zero and covariance Σ_u ; a structural impact matrix D with $\Sigma_u = DD'$ maps them to orthonormal structural shocks through $u_t = D\varepsilon_t$, $\mathbb{E}[\varepsilon_t \varepsilon_t'] = I_K$. Reduced-form moving-average coefficients are $\Phi_j = JA^j J'$ (with A the companion matrix and J the top-block selector), structural ones $\Theta_j = \Phi_j D$, and $R^+ = R'(RR')^{-1}$ is the pseudoinverse of a full-row-rank R .

2 The model and conditional forecasts

We work with the reduced-form VAR

$$y_t = \nu + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \quad (1)$$

written in companion form $Y_t = \mu + AY_{t-1} + U_t$ and iterated forward. Taking the conditional expectation kills every future shock, leaving the baseline (unconditional) forecast

$$\mathbb{E}[y_{t+h} \mid \mathcal{F}_t] = \left(\sum_{j=0}^{h-1} JA^j J' \right) \nu + JA^h Y_t, \quad (2)$$

while the deviation of any realized path from this baseline is driven entirely by future shocks,

$$y_{t+h} - \mathbb{E}[y_{t+h} \mid \mathcal{F}_t] = \sum_{j=0}^{h-1} \Phi_j u_{t+h-j} = \sum_{j=0}^{h-1} \Theta_j \varepsilon_{t+h-j}. \quad (3)$$

These steps are standard and Appendix A records them in full; the one object the rest of the paper needs is the stacked structural map. Collecting (3) for $h = 1, \dots, H$ and writing $\varepsilon = [\varepsilon'_{t+1}, \dots, \varepsilon'_{t+H}]'$ for the stack of future structural shocks, the deviation of the whole forecast path from the baseline is a single linear image of those shocks,

$$\Delta y = \mathcal{M} \varepsilon, \quad \mathcal{M} = \begin{bmatrix} \Theta_0 & & & \\ \Theta_1 & \Theta_0 & & \\ \vdots & & \ddots & \\ \Theta_{H-1} & \Theta_{H-2} & \cdots & \Theta_0 \end{bmatrix}, \quad (4)$$

a block-lower-triangular $KH \times KH$ matrix with $\Theta_0 = D$ on the diagonal. Everything below is a linear operation on (4); working in the structural stack, where ε has identity covariance, is what keeps the uncertainty algebra of the next section clean.

A scenario imposes target values on a subset of variables over some horizons. Each imposed cell is one linear equation in ε , obtained by selecting the corresponding row of $\mathcal{M}$, so a scenario is the linear restriction

$$R \varepsilon = r, \quad R = \mathcal{M}_{[\text{constrained rows}], :}, \quad r = (\text{target} - \text{baseline})_{[\text{constrained rows}]}, \quad (5)$$

with $r \in \mathbb{R}^m$ collecting the m constrained cells. Because a scenario fixes far fewer cells than there are free shocks, (5) is underdetermined; Waggoner and Zha, after Doan, Litterman and Sims, select the minimum-norm solution

$$\hat{\varepsilon} = \arg \min_{\varepsilon} \frac{1}{2} \varepsilon' \varepsilon \quad \text{s.t.} \quad R \varepsilon = r = R'(RR')^{-1}r = R^+ r, \quad (6)$$

and the conditional forecast is $\Delta y = \mathcal{M} \hat{\varepsilon}$ added back to the baseline. Two points recur below. First, minimizing $\varepsilon' \varepsilon$ minimizes the sum of squared *structural* shocks, which is the Mahalanobis norm $u' \Sigma_u^{-1} u$ of the reduced-form innovations and, under $\varepsilon \sim N(0, I)$, picks out the mode of the shocks consistent with the scenario; minimizing the unweighted reduced-form norm instead would load implausibly large shocks, in standard-deviation terms, onto low-variance equations and distort every unconstrained path, so the distinction is substantive rather than cosmetic. Second, since $\mathbb{E} \|\varepsilon\|^2 = KH$, the ratio $\|\hat{\varepsilon}\|/\sqrt{KH}$ is a readable gauge of how demanding a scenario is—near one for an ordinary draw, several times that for a scenario the model regards as close to impossible—and the dashboard reports it as a traffic light on every exercise.

3 Scenarios and their uncertainty

The restriction (5) is agnostic about how a target path is chosen. In practice a scenario is described in economic terms and translated into the VAR's transformed units: a variable in levels (the unemployment or policy rate) is constrained directly, while a variable in log-differences (oil, prices, housing, employment, the exchange rate) has its authored *level* path $\ell_{1:H}$ mapped by

$$\text{level variable: } r_h^{\text{unit}} = \ell_h, \quad \text{log-diff variable: } r_h^{\text{unit}} = \log \ell_h - \log \ell_{h-1}, \quad (7)$$

with ℓ_0 the last observed level. A small vocabulary of shapes—a *ramp* to a target and hold, a flat *hold*, a step–hold–ease *policy* rate path, a *replay* of a variable’s own historical growth from a chosen date, and a *pulse* that ramps, holds, and is then released—covers the common cases; the constructors are spelled out in Appendix B.2.

A scenario, however, is not a single line. The constraint fixes only m linear combinations of the KH -dimensional shock vector and leaves the rest free, so the conditional forecast of every unconstrained variable has a genuine predictive distribution. Holding the parameters fixed, conditioning the prior $\varepsilon \sim N(0, I)$ on $R\varepsilon = r$ gives another Gaussian, supported on the constraint subspace,

$$\varepsilon \mid R\varepsilon = r \sim N(\hat{\varepsilon}, P), \quad \hat{\varepsilon} = R^+r, \quad P = I - R^+R, \quad (8)$$

where P is the orthogonal projector onto the null space of R ; propagating through (4), the conditional forecast is $\Delta y \sim N(\mathcal{M}\hat{\varepsilon}, \mathcal{M}P\mathcal{M}')$,

$$\text{Var}(\Delta y \mid R\varepsilon = r) = \mathcal{M}P\mathcal{M}'. \quad (9)$$

Because $RP = 0$, the rows of $\mathcal{M}P\mathcal{M}'$ that belong to constrained cells vanish: the fan chart pinches to a point on the variable and horizons the scenario pins, and opens up only where the model is free. This *shock* band is closed form—no simulation is needed—which is what lets the dashboard draw it live as the user edits a scenario. Since the display transforms are nonlinear (levels are reconstructed as $\ell_0 \exp(\text{cumsum})$, growth rates as year-over-year differences), bands are read as sample quantiles of transformed draws from (8), not as transforms of Gaussian quantiles; Appendix B.5 gives the argument.

The estimated parameters carry their own uncertainty, and it cannot be bolted on afterward: the coefficients enter R , r , and $\mathcal{M}$ themselves, so a different draw of the VAR implies a different constraint system and a different conditional mean, and the scenario must be re-solved inside each draw. Drawing from the normal–inverse–Wishart posterior when a shrinkage prior is used, or from a bias-adjusted residual bootstrap otherwise, and taking quantiles of the transformed conditional draws across parameter draws yields the *parameter* band; combined with the shock band it gives the reported fan (Appendix B.6). One reading caveat is worth stating in the main text: the point baseline from the OLS estimate is not the centre of the bootstrap distribution, because the bias adjustment and the rejection of explosive draws shift the per-draw baseline, so the clean object for “is this scenario distinguishable from the baseline?” is the scenario-minus-baseline band computed draw by draw, which nets out the shift.

Partial-horizon conditions come for free: leaving a horizon out of (5) leaves it free, and the pulse type does exactly this, constraining a plateau and releasing everything after it. For a log-difference variable, pinning months $1, \dots, n$ pins the *level* at month n and lets the model supply the persistence, so “oil spikes for six months and then normalizes” becomes a statement the model completes rather than a plateau the analyst imposes. Figure 1 is such a released oil pulse. Over the six constrained months the effect is imposed, but the tail is the model’s own answer: headline inflation climbs from about 2% to a peak near 3.8% in 2027 and then eases back toward 2.5%, housing inflation recovers from mildly negative toward zero, and unemployment and the policy rate barely stir. The bands widen steadily, and by 2028 the 90% interval for inflation runs from roughly zero to 5%—a reminder that a six-month spike says little about the medium term with any confidence, which is exactly what a risk exercise should make visible.

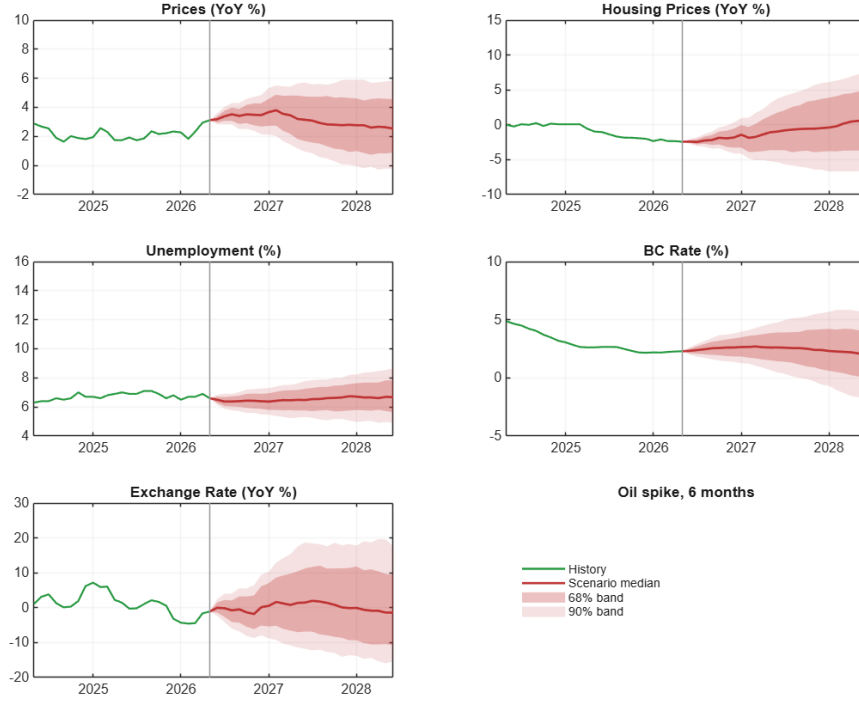


Figure 1: A temporary (“pulse”) oil scenario: oil is constrained for six months and then released, so the tail is the model’s own response. Green is history, red the scenario median, and the shaded bands the 68% and 90% intervals combining shock and parameter uncertainty.

Constraining several variables at once merely adds rows to R and r , so genuinely multivariate exercises are as cheap as univariate ones. This is what makes qualitatively new scenarios reachable: constraining oil up and unemployment up together forces the system onto a supply-shock path that neither constraint produces alone; holding activity down while the policy rate is fixed expresses a “recession with no policy response” as a joint condition; and a tariff shock, which hits both foreign activity and the exchange rate, must constrain the two together, since constraining only one lets the model offset it through the other. Figure 2 shows the stagflation case. Unemployment is held at 8%, visible as the flat median with essentially no band—the zero-width pinch of (9) made concrete—while oil is pushed up; inflation rises briefly toward 3% and then drifts below 2% as the demand drag dominates, and the policy rate falls through the window with a wide band. The only quantity that moves against the analyst is feasibility: a joint condition is far more demanding than either leg, and $\|\hat{\epsilon}\|$ grows accordingly.

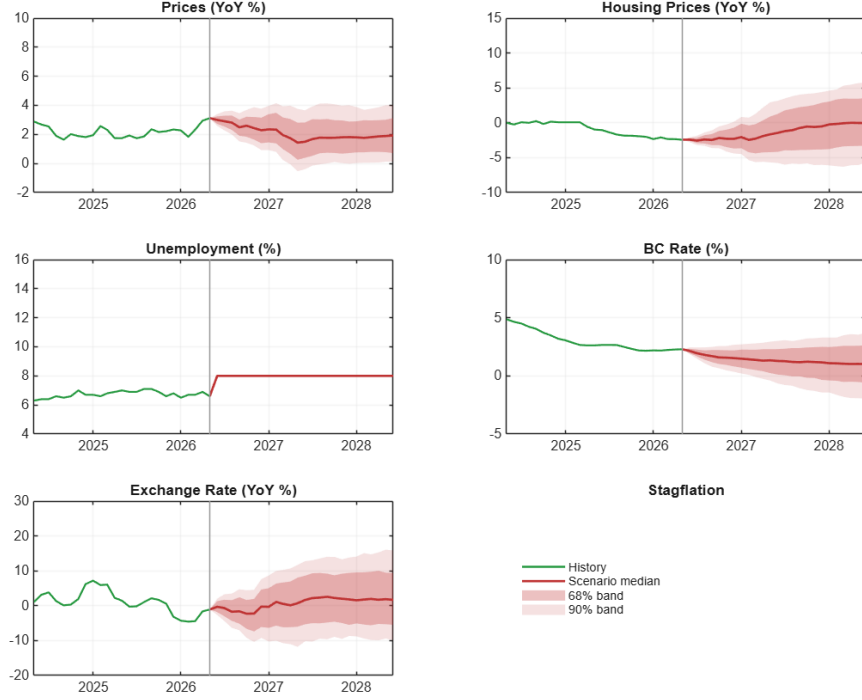


Figure 2: A joint scenario (stagflation): oil and unemployment are constrained upward together. The unemployment panel shows the constrained path as a flat median with no band—the constrained variable’s uncertainty is zero by construction—while the free variables fan out.

A hard condition pins the constrained variable to an exact number, but “oil around \$120” is more honestly a distribution than a point. Writing the restriction as a noisy measurement $r = R\epsilon + e$ with $e \sim N(0, \Omega)$,

$$r = R\epsilon + e, \quad \Omega = \text{diag}(\sigma^2), \quad (10)$$

and updating the prior in the usual Gaussian way gives the soft posterior

$$\epsilon \mid r \sim N(R'(RR' + \Omega)^{-1}r, \quad I - R'(RR' + \Omega)^{-1}R), \quad (11)$$

which returns the hard solution as $\Omega \rightarrow 0$ and otherwise gives the constrained variable a nondegenerate band of its own (Appendix B.4). Softening the same oil pulse, Figure 3, barely moves the medians of the reported variables and only modestly widens their near-term fans; the substantive change is on the constrained variable itself, whose pinned line becomes a band—the more truthful object for something labelled a risk scenario. Softness is set per horizon, so a path can be pinned tightly early and loosely later.

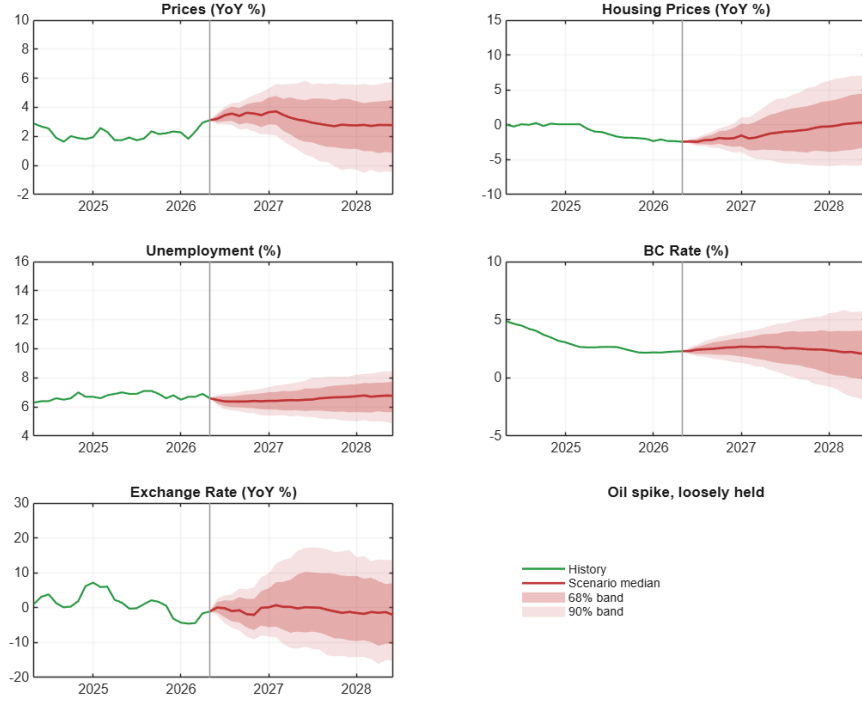


Figure 3: A soft oil condition: conditioning on a distribution rather than an exact path. On the reported variables it is close to the hard pulse of Figure 1; the difference is that the constrained variable itself now carries uncertainty.

4 Historical counterfactuals

Nothing in the algebra confines it to the forecast horizon. Run the same machinery over a window the data actually realized and it answers historical “what if” questions. Over such a window from an origin t_0 , write the outcome as the origin baseline plus the realized innovations, $y_{\text{actual}} = F_{\text{base}} + \mathcal{M}\epsilon_{\text{actual}}$, so that on the constrained rows $R\epsilon_{\text{actual}} = r_{\text{actual}}$ is the realized deviation from the baseline. A counterfactual imposes a different constrained path r_{cf} while changing as little else as possible,

$$\min_{\delta} \|\delta\| \quad \text{s.t.} \quad R(\epsilon_{\text{actual}} + \delta) = r_{\text{cf}} \implies \delta = R^+(r_{\text{cf}} - r_{\text{actual}}), \quad (12)$$

so that

$$y_{\text{cf}} = y_{\text{actual}} + \mathcal{M}R^+(r_{\text{cf}} - r_{\text{actual}}). \quad (13)$$

The baseline cancels: the counterfactual is a perturbation of *history*, not of a forecast, so it keeps every realized shock the constraint does not require changing. A pure policy experiment should move history only through the monetary-policy shock, which is imposed by restricting δ to the columns of R carrying that shock before solving (12) (Appendix B.7). Figure 4 runs precisely this exercise: had the Bank held the policy rate 100 basis points higher through 2022–23, the model puts the price level about 0.6 percentage points lower and the unemployment rate about 0.75 points higher by the end of the window—the shaded wedge between the actual and counterfactual paths—a disinflation bought with labour-market slack, and the expected sign. Because the change

is confined to the policy shock, the 2022 commodity and supply shocks remain in both paths and only the monetary leg differs. Two caveats travel with any such number. The reduced-form VAR is assumed invariant to the policy change contemplated, which is exactly what the Lucas critique denies, so the exercise is informative about the model’s internal propagation, not about what the economy would truly have done; and the size of the required perturbation, $\|\delta\|/\sqrt{KH}$, gauges how far the counterfactual reaches beyond historical experience. Both are surfaced in the dashboard.

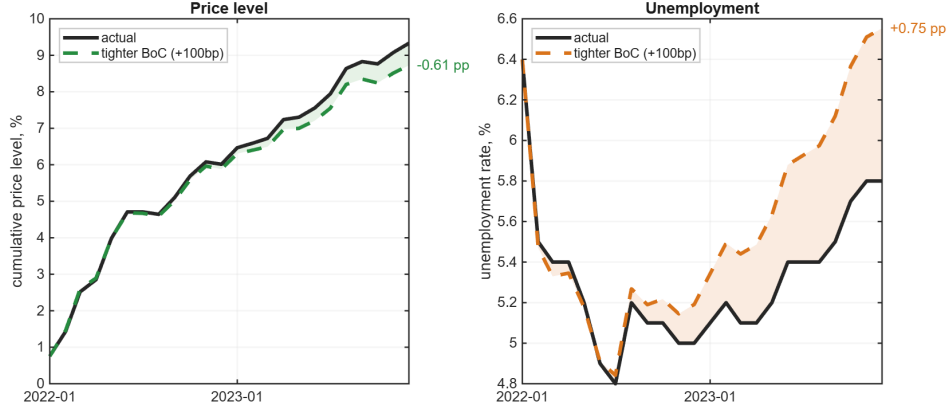


Figure 4: A historical counterfactual: the price level and unemployment over 2022–23 under a 100bp-tighter policy path (dashed) against the realized data (solid). The end-of-window gaps are annotated; the perturbation is confined to the policy shock.

5 Identification

Everything above is conditional on an identification of the impact matrix D . We use the recursive scheme of the source paper, $D = \text{chol}(\Sigma_u)$ lower triangular, which is a set of *timing restrictions*: because D is lower triangular, a shock ordered k moves variables $1, \dots, k-1$ only with a lag and variables $k, \dots, K$ within the period. The ordering therefore encodes a contemporaneous causal chain, and the one used here—global and commodity variables first, then the domestic price block, then activity and the policy rate, with forward-looking financial variables last—reads as a set of economic priors: block exogeneity of a small open economy to world conditions within the month, prices that are sticky relative to the activity and policy block, and asset prices that respond to everything contemporaneously. These are assumptions of convenience as much as of theory; they are not tested by the data, and every object that is expressed in terms of *named* structural shocks inherits them.

It is worth being precise about which results depend on this choice, because it is less than one might expect. The Waggoner–Zha conditional forecast minimises the Mahalanobis norm $u'\Sigma_u^{-1}u$ of the reduced-form innovations, and both the reduced-form dynamics and Σ_u are invariant to how one factors $\Sigma_u = DD'$. The scenario means of Section 3, and the shock bands built from the conditional covariance, are therefore *invariant* to the Cholesky ordering: permuting the recursive order, or indeed replacing the Cholesky factor by any other valid D , leaves every conditional forecast and every fan chart unchanged. What does depend on the identification is anything attributed to a particular structural shock: the impulse responses $\Theta_j = \Phi_j D$, the forecast-error variance decomposition, and—among the exercises developed here—the counterfactual that is *confined to a chosen shock*, such as the tighter-policy experiment of Section 4, whose confinement singles out

the policy column of D . Identification thus matters exactly where a result is read as “the effect of shock ℓ ,” and is immaterial where a scenario is expressed purely as a path for observed variables.

That distinction also bounds the cost of the recursive scheme’s known weaknesses. Recursive timing restrictions are prone to the familiar puzzles—a contractionary monetary shock that appears to raise prices, or an activity response of the wrong sign—and the results can be sensitive to the ordering, which is hard to defend at the monthly frequency for variables that in truth respond to one another without delay. Because the identification is a single input, D , the toolkit is agnostic to how it is obtained, and a natural line of further work is to replace the Cholesky factor with an identification that rests on economics rather than timing. Several are under review: sign restrictions on the impulse responses, which set-identify the shocks from theory-consistent comovements (Uhlig, 2005; Rubio-Ramírez, Waggoner and Zha, 2010); external- instrument or proxy-SVAR identification, which uses an outside series correlated with a single structural shock—high-frequency policy surprises for the monetary shock, for instance (Mertens and Ravn, 2013; Gertler and Karadi, 2015; Stock and Watson, 2018); identification through heteroskedasticity, which exploits changes in the shock variances across regimes (Rigobon, 2003); and narrative restrictions that discipline the shocks with information about specific historical episodes (Antolín-Díaz and Rubio-Ramírez, 2018). Each of these delivers a different D (or a set of admissible D ’s), and each can be substituted into the pipeline without touching the scenario machinery: the impulse responses, the variance decomposition, and the shock-confined counterfactuals would update, while the scenario forecasts and their bands would not move at all.

6 Implementation and the dashboard

The methods are delivered as software, and the software is part of the contribution. The estimation runs from a cached panel with no network access, so a run is deterministic and archivable, and a one-time online step freezes the data into that cache. A regression suite checks the data-independent invariants any correct refactor must preserve—that Σ_u is symmetric positive definite, that DD' recovers it, that the impact response equals D , that forecast-error-variance shares sum to one, that hard conditioning hits its constraint to numerical zero, and that a counterfactual with $r_{cf} = r_{actual}$ reproduces history exactly—alongside a golden-value snapshot that catches numerical drift. A report generator collects every figure and the variance-decomposition table into a single self-contained HTML file.

The interactive dashboard is a single self-contained HTML page that does all of its computation in the browser from the exported model objects: the structural responses Θ_j , the impact matrix D , the baseline, the companion matrix, and the transformed history. An impulse-response explorer rescales and recumulates responses on the fly; a scenario builder lets the user assemble hard or soft, single or joint, full or partial-horizon constraints and watch the conditional forecast—and, on demand, the closed-form shock band of Section 3—recompute as the inputs change, with the in-browser Waggoner–Zha solve validated against the reference implementation to about 10^{-8} ; a counterfactual tab implements (13) with the structural-shock confinement and the feasibility and Lucas-critique readouts; and a grouped gallery holds the archived fan charts. Because the browser recomputes from exported matrices rather than re-estimating, the parameter-uncertainty band remains an offline object; exporting a set of posterior draws to make the full band live is a natural next step.

7 Evaluation and outlook

The apparatus is internally coherent and reproducible, and the figures behave as the theory says they should—the constrained variable pinches, the free variables fan out, soft conditions widen the constrained variable’s own band, and the counterfactual signs are the expected ones. What remains is to show that the conditional forecasts are also *good*. The natural test is a pseudo-real-time evaluation: re-estimate on an expanding window and score the forecasts against realized data, both by point-forecast loss relative to an autoregressive benchmark and by a proper density score computed from predictive quantiles, which would also tell us whether the fan-chart widths are calibrated. The evaluation framework—the benchmark, the error definitions, and the quantile representation of the continuous ranked probability score—is standard and is recorded in [Appendix A](#); the back-testing harness that would populate it is the principal item of future work, and it reuses the same conditional-forecast solve developed here inside a date loop.

To conclude, conditional forecasting is usually presented as a point exercise, but the method is a statement about a distribution of shocks and therefore carries its own uncertainty. Making that uncertainty the object of interest—a fan chart with a closed-form shock component and a simulated parameter component, soft conditions that give the constrained variable a band, joint and partial-horizon scenarios as row operations on one restriction, and historical counterfactuals as minimum-norm perturbations of history—turns a familiar tool into a more honest one, and the interactive delivery puts it in the hands of anyone with a browser. The remaining gap is empirical validation of the forecasts themselves, which the planned out-of-sample evaluation is designed to fill.

A Foundations

This appendix derives, from the reduced-form VAR, the objects that the new results of Appendix B take as given: the companion form and forward recursion, the baseline forecast, the moving-average coefficients and their stacked form $\mathcal{M}$, the Waggoner–Zha minimum-norm solution $\hat{u} = R^+r$, and the structural identification $u = D\varepsilon$. Rather than reproduce the source paper’s results wholesale, we keep only what the extensions build on, and each subsection closes with a pointer (*Used in:*) to where it is used downstream. The source paper’s variance decomposition and out-of-sample metrics are recorded at the end (§A.7) for reference; they are not prerequisites for the new results. The structural-space restriction used in the main text, $R\varepsilon = r$ with stacked map $\mathcal{M}$, is the reduced-form restriction here under $u = D\varepsilon$ (§B.1); the standalone `derivation_companion.tex` carries the same material in the source paper’s own numbering and prose.

A.1 Reduced form and companion form

The starting point is the reduced-form VAR

$$y_t = \nu + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + u_t, \quad \mathbb{E}[u_t] = 0, \quad \mathbb{E}[u_t u_t'] = \Sigma_u, \quad (1)$$

a K -dimensional linear autoregression of order p . “Reduced form” means u_t collects statistical innovations with *no* economic interpretation; the A_i summarise own- and cross-dynamics. Because every equation shares the same right-hand-side regressors, the seemingly-unrelated-regression system collapses to equation-by-equation OLS.

Companion (state-space) form. Stacking the current and $p - 1$ most recent vectors turns (1) into a first-order system,

$$Y_t = \mu + AY_{t-1} + U_t, \quad (2)$$

with

$$Y_t = \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}, \quad \mu = \begin{bmatrix} \nu \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad U_t = \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad A = \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_K & 0 \end{bmatrix}.$$

Reading (2) block-row by block-row: the first block row is exactly (1); block row $i \geq 2$ is the identity $y_{t-i+1} = y_{t-i+1}$, which merely carries lags forward in the state. With the top-block selector $J = [I_K \ 0 \ \cdots \ 0] \in \mathbb{R}^{K \times Kp}$ one has the identities used repeatedly below,

$$y_t = JY_t, \quad \mu = J'\nu, \quad U_t = J'u_t. \quad (\star)$$

Covariance-stationarity requires all eigenvalues of A to lie inside the unit circle.

Used in: everything below; the companion matrix A is exported to the dashboard so the counterfactual baseline (§B.7) can be iterated in the browser from any in-sample origin.

A.2 Forward iteration: the h -step recursion

Iterating (2) forward h periods gives

$$Y_{t+h} = \left(\sum_{j=0}^{h-1} A^j \right) \mu + A^h Y_t + \sum_{j=0}^{h-1} A^j U_{t+h-j}. \quad (3)$$

Derivation (induction on h). The base case $h = 1$ is (2). Assuming (3) at horizon h and applying (2) once more, $Y_{t+h+1} = \mu + AY_{t+h} + U_{t+h+1}$, substitution gives

$$Y_{t+h+1} = \underbrace{\mu + \sum_{j=0}^{h-1} A^{j+1} \mu + A^{h+1} Y_t}_{=\sum_{j=0}^h A^j \mu} + \underbrace{\sum_{j=0}^{h-1} A^{j+1} U_{t+h-j} + U_{t+h+1}}_{=\sum_{j=0}^h A^j U_{t+h+1-j}},$$

where the last bracket re-indexes $j \mapsto j+1$ (turning the sum into $\sum_{j=1}^h A^j U_{t+h+1-j}$) and the stray U_{t+h+1} supplies the missing $j=0$ term. This is (3) at $h+1$.

Used in: the baseline (§A.3) is its conditional mean; the moving-average map (§A.4) is its stochastic part.

A.3 The baseline forecast

Taking $\mathbb{E}[\cdot \mid \mathcal{F}_t]$ of (3) annihilates every future shock, since U_{t+h-j} is dated $t+h-j \geq t+1$ for $j = 0, \dots, h-1$ and hence unforecastable, $\mathbb{E}[U_{t+h-j} \mid \mathcal{F}_t] = 0$. Premultiplying by J and using $\mu = J'\nu$ from (★),

$$\mathbb{E}[y_{t+h} \mid \mathcal{F}_t] = \left(\sum_{j=0}^{h-1} J A^j J' \right) \nu + J A^h Y_t. \quad (4)$$

Under quadratic loss this conditional expectation is the optimal point forecast; it is the *baseline scenario* against which every conditional forecast and every counterfactual is measured.

Used in: the constraint right-hand side $r = (\text{target} - \text{baseline})$ of the main text (5) and the counterfactual origin baseline F_{base} (§B.7).

A.4 Forecast error, impulse responses, and the stacked map

Subtracting (4) from (3) leaves only the shock term; projecting onto the top block with $U_{t+h-j} = J' u_{t+h-j}$ gives the deviation of any realised path from the baseline,

$$y_{t+h} - \mathbb{E}[y_{t+h} \mid \mathcal{F}_t] = \sum_{j=0}^{h-1} J A^j J' u_{t+h-j} = \sum_{j=0}^{h-1} \Phi_j u_{t+h-j}, \quad \Phi_j := J A^j J'. \quad (5)$$

The Φ_j are the reduced-form moving-average (impulse-response) coefficients, with $\Phi_0 = I_K$. **Economic reading:** forcing a variable onto a scenario path (left-hand side $\neq 0$) is *equivalent* to imposing a particular realisation of future innovations (right-hand side); the scenario cannot happen “for free,” and the same innovations move every other variable through Φ_j .

The stacked map. Collect (5) for $h = 1, \dots, H$. Writing $\Delta y = [(y_{t+1} - \mathbb{E})', \dots, (y_{t+H} - \mathbb{E})']' \in \mathbb{R}^{KH}$ and stacking the innovations conformably, the deviation is a single linear image of the future shocks,

$$\Delta y = \mathcal{M}_u u, \quad \mathcal{M}_u = \begin{bmatrix} \Phi_0 & & & \\ \Phi_1 & \Phi_0 & & \\ \vdots & & \ddots & \\ \Phi_{H-1} & \Phi_{H-2} & \cdots & \Phi_0 \end{bmatrix}, \quad (5')$$

a block-lower-triangular $KH \times KH$ matrix (block $(h, s) = \Phi_{h-s}$ for $s \leq h$, zero otherwise). Once a structural impact matrix D is identified (§A.6), the same construction with $\Theta_j := \Phi_j D$ in place of Φ_j gives the *structural* stacked map $\mathcal{M}$ of the main text (4), whose diagonal block is $\Theta_0 = D$.

Used in: $\mathcal{M}$ is the object of the paper. The restriction rows R are selected rows of $\mathcal{M}$; the propagation $\Delta y = \mathcal{M}\varepsilon$ delivers every conditional forecast, band (§B.5, §B.6), and counterfactual (§B.7). The coefficients Θ_j are exported to the dashboard, which rebuilds $\mathcal{M}$ in the browser.

A.5 The conditioning restriction and the minimum-norm shock path

A scenario fixes target values for a subset of variables over some horizons. Reading the corresponding rows off (5') yields a linear system in the stacked future innovations,

$$Ru = r, \quad (5'')$$

where $r \in \mathbb{R}^m$ is the gap between the scenario values and the baseline forecasts on the m constrained cells and $R \in \mathbb{R}^{m \times KH}$ collects the associated rows of the MA coefficients. Because a scenario fixes far fewer cells than there are free shocks ($m < KH$), (5'') is underdetermined; WZ, after Doan–Litterman–Sims, select the minimum-norm solution.

Derivation (constrained minimisation). Solve $\min_u \frac{1}{2}u'u$ s.t. $Ru = r$. The Lagrangian $\mathcal{L} = \frac{1}{2}u'u + \lambda'(r - Ru)$ has first-order conditions $u = R'\lambda$ and $Ru = r$; substituting, $RR'\lambda = r$, so $\lambda = (RR')^{-1}r$ (full row rank m) and

$$\hat{u} = R'(RR')^{-1}r = R^+ r. \quad (5''')$$

Adding $\hat{u}$ back through (5') delivers the conditional forecast for every variable.

Remark 1 (dimensions of the inverse). With R of size $m \times KH$ and $m < KH$, RR' ($m \times m$) is invertible while $R'R$ ($KH \times KH$) is singular. A source-paper display that writes $\hat{u} = R'(R'R)^{-1}r$ should read $(RR')^{-1}$ as in (5'''); the two coincide only under the transposed storage convention for R . The substance is unchanged.

Remark 2 (“most likely” shocks). If $u \sim N(0, I_{KH})$ then (5''') is the mode (equivalently the mean) of $u \mid Ru = r$: minimising $u'u$ maximises the Gaussian density. This is why $\hat{u}$ is the most likely combination of shocks consistent with the scenario — the reading the uncertainty derivations of Appendix B make precise.

Used in: (5''') is the object every new derivation extends — soft conditioning replaces $(RR')^{-1}$ by $(RR' + \Omega)^{-1}$ (§B.4); the shock band conditions the Gaussian around $\hat{\varepsilon}$ (§B.5); the counterfactual perturbs a realised $\hat{u}$ (§B.7). The structural-space version $\hat{\varepsilon} = R^+ r$ used in the main text is (5''') with $u = D\varepsilon$ (§B.1).

A.6 Structural identification

The reduced-form innovation is decomposed as $u_t = D\varepsilon_t$ with orthonormal structural shocks,

$$\mathbb{E}[\varepsilon_t] = 0, \quad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I_K \implies \Sigma_u = DD'. \quad (6)$$

Order condition and Cholesky. A symmetric Σ_u has $K(K+1)/2$ distinct entries while D has K^2 ; the equations $\Sigma_u = DD'$ fall short by $K^2 - K(K+1)/2 = K(K-1)/2$ restrictions. Requiring D lower-triangular zeros out exactly its $K(K-1)/2$ upper entries, closing the gap, and with a positive diagonal the solution is the unique Cholesky factor $D = \text{chol}(\hat{\Sigma}_u)$, with $\hat{\Sigma}_u = \sum_t \hat{u}_t \hat{u}_t' / (T - Kp - 1)$. A lower-triangular D means shock k moves variables $1, \dots, k-1$ only with a lag but $k, \dots, K$ contemporaneously, so the recursive ordering of y_t is material (global variables first, sticky prices ahead of the activity/policy block, forward-looking financial variables last). The structural impulse responses are $\Theta_j = \Phi_j D$.

Used in: $D = \Theta_0$ is the diagonal block of $\mathcal{M}$ and the change of variables $u = D\varepsilon$ that turns the WZ norm into the Mahalanobis norm (§B.1); the whole apparatus is identification-agnostic — replace the Cholesky D by a sign-restricted or instrument-based D and every formula in Appendix B is unchanged.

A.7 Reference: variance decomposition and out-of-sample metrics

The following source-paper results are recorded for completeness; they are *not* prerequisites for the new derivations of Appendix B.

Forecast-error variance decomposition. With the structural forecast error $y_{t+h} - \mathbb{E}y_{t+h} = \sum_{j=0}^{h-1} \Theta_j \varepsilon_{t+h-j}$ and orthonormal ε , the mean squared prediction error and its per-shock split are

$$\text{MSPE}(h) = \sum_{j=0}^{h-1} \Theta_j \Theta_j', \quad \text{MSPE}_{k,l}(h) = \sum_{j=0}^{h-1} \theta_{k,l,j}^2, \quad (7', 7'')$$

where $\theta_{k,l,j} = [\Theta_j]_{kl}$; the reported share is $\text{MSPE}_{k,l}(h) / \sum_l \text{MSPE}_{k,l}(h)$, and orthogonality is what makes the shares sum to one across l .

Out-of-sample evaluation (planned exercise). Point accuracy uses an $\text{AR}(p)$ benchmark — (1) with diagonal A_i — and h -step errors $e_{k,t}^{(h,m)}$ aggregated into

$$\text{RMSE}(k, h, m) = \sqrt{\frac{1}{\bar{t} - \underline{t}} \sum_t (e_{k,t}^{(h,m)})^2}, \quad \text{MAE}(k, h, m) = \frac{1}{\bar{t} - \underline{t}} \sum_t |e_{k,t}^{(h,m)}|, \quad (9, 10)$$

each reported relative to the benchmark (a ratio below one favours the VAR). Density accuracy uses the continuous ranked probability score in its quantile (“pinball”) form: with $\rho_\tau(u) = u(\tau - \mathbb{I}(u < 0))$ and predictive quantiles $\hat{q}_{\tau,k,t}^{(h,m)}$ on a grid $\tau_1 < \dots < \tau_N$,

$$\text{CRPS}(k, h, m) = \frac{2}{N-1} \sum_{n=1}^N \left(\frac{1}{\bar{t} - \underline{t}} \sum_t \rho_{\tau_n}(y_{k,t} - \hat{q}_{\tau_n,k,t}^{(h,m)}) \right), \quad (11)$$

the trapezoidal discretisation of $\text{CRPS}(F, y) = 2 \int_0^1 \rho_\tau(y - F^{-1}(\tau)) d\tau$, which needs only the predictive quantiles, not the full density. These metrics drive the planned backtesting harness (§7).

Summary of the foundation chain

$$\underbrace{(1) \Rightarrow (2)}_{\text{companion}} \Rightarrow \underbrace{(3)}_{\text{iterate}} \Rightarrow \begin{cases} \mathbb{E}[\cdot | \mathcal{F}_t] \Rightarrow (4) \text{ baseline} \\ \text{deviation} \Rightarrow (5), (5') \Rightarrow \mathcal{M} \text{ (stacked map)} \\ Ru = r \Rightarrow \hat{u} = R^+ r \text{ (min-norm, WZ)} \\ u = D\varepsilon, \Sigma_u = DD' \Rightarrow \Theta_j = \Phi_j D \text{ (identification)} \end{cases}$$

Appendix B builds on these objects: it adds a distribution around $\hat{u}$ (bands), a noisy version of $Ru = r$ (soft conditions), extra rows (joint / partial-horizon scenarios), and a perturbation of a realised $\hat{u}$ (counterfactuals).

B New derivations

B.1 Structural versus reduced-form norm

Reduced-form innovations satisfy $u = D\varepsilon$, stacked as $\Delta y = \mathcal{M}_u u = \mathcal{M}\varepsilon$ with $\mathcal{M}_u$ the reduced-form stacked map (blocks Φ_j) and $\mathcal{M}$ its structural counterpart (blocks $\Theta_j = \Phi_j D$). The reduced-form restriction $R_u u = r$ and the structural one $R\varepsilon = r$ have $R = R_u D_{\text{blk}}$, $u = D_{\text{blk}}\varepsilon$, where $D_{\text{blk}} = I_H \otimes D$. Minimizing $\varepsilon'\varepsilon$ subject to $R\varepsilon = r$ is minimizing $u'D_{\text{blk}}^{-\top}D_{\text{blk}}^{-1}u = u'(I_H \otimes \Sigma_u^{-1})u$, the Mahalanobis norm. Under $\varepsilon \sim N(0, I)$ the density is $\propto \exp(-\frac{1}{2}\varepsilon'\varepsilon)$, so (6) is the mode of $\varepsilon \mid R\varepsilon = r$, justifying “most likely shocks.” The two norms differ in every unconstrained variable; the structural (Mahalanobis) norm is the correct one.

B.2 The scenario-to-units map

Let ℓ_0 be the last observed level of the variable and x_0 its value. The constructors used to author a path (all in *level* space, then mapped by (7)) are:

- *ramp* to $\bar{\ell}$ over n months: $\ell_h = x_0 + (\bar{\ell} - x_0) \min(h/n, 1)$;
- *hold* at a : $\ell_h = a$ for all h ;
- *pulse* to $\bar{\ell}$ over n , hold k : the ramp-hold on $h \leq n+k$, and $\ell_h = \text{NaN}$ (free) for $h > n+k$;
- *policy*: r_h steps up by s each month to a peak $\bar{r}$, holds k months, then eases by s to a floor $\underline{r}$ (a level-variable path);
- *replay* from date τ for n months: copy the variable’s own realized growth (for a log-diff variable) or level (for a level variable) starting just after τ , then free.

A *window* $[h_1, h_2]$ sets $r_h^{\text{unit}} = \text{NaN}$ outside the window; NaN cells contribute no row to (5). For a log-difference variable, note the asymmetry: constraining months $1, \dots, n$ pins the level at month n , whereas a window that does not start at $h = 1$ constrains growth rates without pinning the level they build on.

B.3 Joint and partial-horizon constraints

For M constrained variables with per-variable, per-horizon targets, the row set of (5) is

$$\{ (h, v) : \text{target}(v, h) \text{ is not NaN} \}, \quad \text{row}(h, v) = (h - 1)K + v,$$

so R is $\mathcal{M}$ restricted to those rows and r the corresponding target-minus-baseline entries. Nothing else changes: the solve (6), the soft update (11), and the bands (8) all take this R as input. Duplicated (variable, horizon) cells are disallowed (a variable cannot be constrained twice at the same horizon).

B.4 Soft conditioning (Gaussian update)

Take the prior $\varepsilon \sim N(0, I_{KH})$ and the noisy measurement (10), $r = R\varepsilon + e$, $e \sim N(0, \Omega)$, $e \perp \varepsilon$. The joint law is Gaussian with

$$\text{Cov}(\varepsilon, r) = R', \quad \text{Var}(r) = RR' + \Omega, \quad \mathbb{E}[\varepsilon] = 0, \quad \mathbb{E}[r] = 0.$$

The standard Gaussian conditioning formula $\mathbb{E}[\varepsilon \mid r] = \text{Cov}(\varepsilon, r)\text{Var}(r)^{-1}r$ and $\text{Var}[\varepsilon \mid r] = \text{Var}(\varepsilon) - \text{Cov}(\varepsilon, r)\text{Var}(r)^{-1}\text{Cov}(r, \varepsilon)$ give (11):

$$\hat{\varepsilon}_{\text{soft}} = R'(RR' + \Omega)^{-1}r, \quad P_{\text{soft}} = I - R'(RR' + \Omega)^{-1}R.$$

As $\Omega \rightarrow 0$, $R'(RR')^{-1} = R^+$ and $P_{\text{soft}} \rightarrow I - R^+R = P$, the hard-condition projector. A posterior draw is generated without forming P_{soft} : with $z \sim N(0, I)$ and $e \sim N(0, \Omega)$,

$$\varepsilon^{\text{draw}} = \hat{\varepsilon}_{\text{soft}} + (z - R'(RR' + \Omega)^{-1}(Rz + e)),$$

whose mean is $\hat{\varepsilon}_{\text{soft}}$ and whose covariance is exactly P_{soft} ; $\Omega = 0$ collapses it to the hard projector draw of §B.5.

B.5 Shock-uncertainty band

Condition $\varepsilon \sim N(0, I)$ on $R\varepsilon = r$ (hard). Decompose ε into its component in $\text{row}(R)$ and its component in $\ker R$. The row-space component is fixed by the constraint at R^+r ; the kernel component is unconstrained and retains its prior law. Hence $\varepsilon \mid R\varepsilon = r \sim N(\hat{\varepsilon}, P)$ with $\hat{\varepsilon} = R^+r$ and $P = I - R^+R$ the orthogonal projector onto $\ker R$ (idempotent, symmetric, $RP = 0$). A draw is $\hat{\varepsilon} + Pz = \hat{\varepsilon} + (z - R^+(Rz))$, applied through the shared factors of R so P is never formed. Propagating, $\Delta y = \mathcal{M}\varepsilon \sim N(\mathcal{M}\hat{\varepsilon}, \mathcal{M}P\mathcal{M}')$, which is (9). Since $RP = 0$, the constrained rows of $\mathcal{M}P\mathcal{M}'$ vanish and the constrained cells have zero predictive variance—the pinch in the fan chart. Because the display transform (level reconstruction $\ell_0 \exp(\sum)$, year-over-year growth) is nonlinear, bands are read as sample quantiles of the transformed draws, not as transforms of the Gaussian quantiles.

B.6 Parameter-uncertainty band

Let $\{(A^{(d)}, \Sigma_u^{(d)})\}_{d=1}^N$ be draws of the VAR parameters. For each: form $D^{(d)} = \text{chol}(\Sigma_u^{(d)})$, the structural map $\mathcal{M}^{(d)}$ (blocks $\Theta_j^{(d)} = \Phi_j^{(d)}D^{(d)}$), the baseline $F_{\text{base}}^{(d)}$, then rebuild $R^{(d)}, r^{(d)}$ and solve $\hat{\varepsilon}^{(d)} = (R^{(d)})^+r^{(d)}$. The per-draw conditional *mean* is $F_{\text{base}}^{(d)} + \mathcal{M}^{(d)}\hat{\varepsilon}^{(d)}$ and a full predictive draw adds a shock draw $\mathcal{M}^{(d)}(z - \text{proj})$ from §B.5 at draw d 's parameters. Reported quantities are quantiles, across d , of the transformed conditional draws (*cond*); of the baseline draws (*base*); and of the scenario-minus-baseline means $\mathcal{M}^{(d)}\hat{\varepsilon}^{(d)}$ (*diff*), which isolate parameter uncertainty because with parameters fixed the difference is deterministic.

Draw mechanism. With a Minnesota prior, draws are from the normal-inverse-Wishart posterior (rejecting explosive companions); otherwise a residual bootstrap regenerates a sample and re-estimates. For the bootstrap, the data-generating companion is bias-adjusted following Kilian (1998): estimate the mean bias by a first bootstrap pass, subtract it, and shrink the adjustment toward zero until the companion is nonexplosive; residuals are resampled only from months not activated by an impulse dummy (whose residuals are zero by construction), so those months are not smuggled back in. Explosive draws in the main pass are redrawn. The OLS baseline is *not* the bootstrap median (the bias adjustment and rejection shift the per-draw baseline), which is why the *diff* object, computed draw by draw, is the correct basis for “distinguishable from baseline.”

B.7 Historical counterfactual

Fix an origin t_0 within the sample and a window of length H . The realized data over the window equal the origin baseline plus the realized structural innovations, $y_{\text{actual}} = F_{\text{base}} + \mathcal{M}\varepsilon_{\text{actual}}$, so on

the constrained rows $R\epsilon_{\text{actual}} = r_{\text{actual}}$ with $r_{\text{actual}} = (y_{\text{actual}} - F_{\text{base}})_{[\text{rows}]}$. Impose a counterfactual target r_{cf} on those rows and keep every other realized shock by minimizing the change:

$$\min_{\delta} \|\delta\| \quad \text{s.t.} \quad R(\epsilon_{\text{actual}} + \delta) = r_{\text{cf}} \Rightarrow R\delta = r_{\text{cf}} - r_{\text{actual}} \Rightarrow \delta = R^+(r_{\text{cf}} - r_{\text{actual}}),$$

by the same Lagrangian argument as (6). Then $y_{\text{cf}} = y_{\text{actual}} + \mathcal{M}\delta$, i.e. (13); the baseline cancels because it is common to actual and counterfactual. To confine the change to a subset S of structural shocks, restrict δ to the columns of R indexed by S across horizons, $R_S = R_{\cdot, [S]}$, solve $\delta_S = (R_S)^+(r_{\text{cf}} - r_{\text{actual}})$, and embed δ_S back into a zero vector. The reported perturbation size is $\|\delta\|/\sqrt{KH}$; a rank-deficient R_S (an infeasible confined target) is handled by the pseudoinverse, which projects the shortfall away. The reduced-form VAR is held fixed throughout—the Lucas-critique assumption that bounds the interpretation.

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